



**RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES**

**B.Sc. (General) Degree in applied Sciences
B.Sc. Second year Semester II Examination – August/September 2021**

CHE 2103 – ANALYTICAL CHEMISTRY

Time: One (01) hour

Answer both questions.

The use of a non-programmable calculator is permitted.

1. a) Round the following numbers to 3 significant figures:
- i) 234,555,359
 - ii) 0.090035
 - iii) 939.25
 - iv) 14090
 - v) 0.008499
- b) Calculate the answer with the absolute uncertainty and percent uncertainty of $1.54 (\pm 0.02) + 2.11 (\pm 0.03) - 0.78 (\pm 0.02)$
- c) Calculate the mean and median for the data given below;
20.1, 19.6, 19.4, 19.5, 19.8 and 20.8
- d) If a given sample of silver and fluorine ideally combine to form 0.6 mol of silver fluoride. What is the percent error if the actual yield is 43 grams?
(molar mass of AgF = 127 g mol^{-1})
- e) The mercury presents in samples of three fish taken from the Nuwara wewa lake was determined by a method based on the absorption of radiation by gaseous elemental mercury. Calculate the pooled estimate of the standard deviation for the method:

Specimen	Number of samples measured	Hg / ppm
1	3	1.80, 1.58, 1.64
2	4	0.96, 0.98, 1.02, 1.10
3	2	3.13, 3.35

- f) Describe the three possible processes of an impurity can be incorporated into a crystal.
- g) Phosphorous in a 1.342 g sample was precipitated as $\text{Ca}_3(\text{PO}_4)_2$ weighing 0.752 g. Express the percentage of phosphorous in the sample expressed as P_4O_{10} in the sample.
(formula weights of $\text{Ca}_3(\text{PO}_4)_2$ and P_4O_{10} are 310 g mol^{-1} and 284 g mol^{-1} respectively).
- h) Write down the mass balanced and the charge balanced equations of a $7.900 \times 10^{-8} \text{ mol dm}^{-3}$ solution of the strong acid HI and calculate the pH of the solution. (pK_w of water at $25^\circ\text{C} = 1.0 \times 10^{-14}$).

(120 marks)

2. Answer Part A or Part B

Part A

- i) Use the following data and calculate the variance and coefficient of variance.
- ii) A manufacturer of a certain thermometer claims that his instrument has a mean lifetime of 20 months. A random sample of 7 such thermometers gave the values of their lifetimes as; 19, 21, 25, 16, 17, 14 and 21. Can you regard the manufacturer's claim to be valid at 1 % level of significant? (values of t for 99% confidence level are:

Degree of freedom, ν	99% confidence level
5	4.032
6	3.707
7	3.500

(50 marks)

- b) A 1.22 g sample of impure lead nitrate was dissolved in 40.0 mL of distilled water and analyzed by gravimetric analysis to determine the percentage of lead by mass present. Sodium sulphate was delivered from a burette to precipitate the lead ions as lead sulphate. The precipitate was filtered, washed and dried before been weighed and had a mass of 1.44 g. Calculate the percentage composition of lead in the original sample.
(O = 16, S = 32 and Pb = 207)

(30 marks)

Part B

- a) Explain the following terms in the formation of the precipitate.
- i) Supersaturation.
 - ii) Nucleation.
- (10 marks)
- b) Calculate the solubility of $\text{Fe}(\text{OH})_3$ in water by systematic approach.
- K_{sp} of $\text{Fe}(\text{OH})_3 = 2 \times 10^{-39} \text{ (mol dm}^{-3}\text{)}^4$
- $K_{\text{w}} = [\text{H}_3\text{O}^+][\text{OH}^-] = 1.00 \times 10^{-14} \text{ (mol dm}^{-3}\text{)}^2$
- (50 marks)
- (c) Compute the volume of $1.20 \times 10^{-1} \text{ mol L}^{-1}$, KOH required to reach the $\text{pH} = 6.27$ of the solution of 100.0 mL, $3.50 \times 10^{-3} \text{ mol L}^{-1}$ diprotic maleic acid.
- (for maleic acid, $\text{pK}_{\text{a1}} = 1.92$ and $\text{pK}_{\text{a2}} = 6.27$).
- (20 marks)

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